

2.

overall balance:

$$A: \overset{10}{n_{A5}} + \overset{0}{n_{A8}} = \overset{0}{n_{A1}} - 2\dot{J}_1 \rightarrow 10 = n_{A1} - 2\dot{J}_1$$

$$B: \overset{2}{n_{B5}} + \overset{0}{n_{B8}} = \overset{0}{n_{B1}} - \dot{J}_1 - \dot{J}_2 \rightarrow 2 = n_{B1} - \dot{J}_1 - \dot{J}_2$$

$$P: \overset{90}{n_{P5}} + \overset{0}{n_{P8}} = \overset{0}{n_{P1}} + \dot{J}_1 - \dot{J}_2 \rightarrow n_{P5} + 90 = \dot{J}_1 + \dot{J}_2$$

$$Q: \overset{30}{n_{Q5}} + \overset{1}{n_{Q8}} = \overset{0}{n_{Q1}} + \dot{J}_2 \rightarrow 30 + 1 = \dot{J}_2 \rightarrow \dot{J}_2 = 31$$

$$n_{Awt} = n_{Am} - 2\dot{J}_1$$

mixer balance:

$$A: n_{A2} + n_{A6} = \overset{0}{n_{A9}} + n_{A1}$$

$$B: n_{B2} + n_{B6} = n_{B9} + n_{B1}$$

$$P: n_{P2} + n_{P6} = \overset{0}{n_{P9}} + \overset{0}{n_{P1}} \rightarrow n_{P2} = n_{P6}$$

$$Q: n_{Q2} + n_{Q6} = \overset{0}{n_{Q9}} + \overset{0}{n_{Q1}} \rightarrow n_{Q2} = n_{Q6}$$

balance separator 2:

$$A: \overset{0}{n_{A8}} + \overset{0}{n_{A9}} = \overset{0}{n_{A1}} \rightarrow n_{A8} = 0$$

$$B: \overset{0}{n_{B8}} + \overset{0}{n_{B9}} = n_{B7} \rightarrow n_{B9} = n_{B7}$$

$$P: \overset{90}{n_{P8}} + \overset{0}{n_{P9}} = n_{P7} \rightarrow n_{P7} = 90$$

$$Q: \overset{1}{n_{Q8}} + \overset{0}{n_{Q9}} = n_{Q7} \rightarrow n_{Q8} = 1$$

$$f_A^{sp} = \frac{\text{mol A consumed}}{\text{mol A fed}} = \frac{2\dot{J}_1}{70 + n_{A2}} = \frac{2\dot{J}_1}{70 + n_{A2}} = .10$$

$$f_B^{sp} = \frac{\text{mol B consumed}}{\text{mol B fed}} = \frac{-\dot{J}_1 - \dot{J}_2}{70 + n_{B2}} = \frac{-\dot{J}_1 - \dot{J}_2}{70 + n_{B2}} = .62$$

separator balance:

$$A: \dot{n}_{A1} + \dot{n}_{A4} = \dot{n}_{A3} \Rightarrow \dot{n}_{A4} = \dot{n}_{A3}$$

$$B: \dot{n}_{B1} + \dot{n}_{B4} = \dot{n}_{B3}$$

$$P: \dot{n}_{P1} + \dot{n}_{P4} = \dot{n}_{P3} \rightarrow \dot{n}_{P1} = \dot{n}_{P3} = 90 \text{ mol/min}$$

$$Q: \dot{n}_{Q1} + \dot{n}_{Q4} = \dot{n}_{Q3} \rightarrow 1 = \dot{n}_{Q3} - \dot{n}_{Q4}$$

splitter balance:

$$A: \dot{n}_{A5} + \dot{n}_{A6} = \dot{n}_{A4}$$

$$x_{A5} \dot{n}_5 + x_{A6} \dot{n}_6 = x_{A4} \dot{n}_4$$

$$x_{A5} = x_{A6} = x_{A4} = x_{Asplit}$$

$$10 \frac{\text{mol}}{\text{min}} = x_{Asplit} (\dot{n}_5 - \dot{n}_6)$$

$$B: \dot{n}_{B5} + \dot{n}_{B6} = \dot{n}_{B4}$$

$$x_{B5} \dot{n}_5 + x_{B6} \dot{n}_6 = x_{B4} \dot{n}_4$$

$$2 = x_{Bsplit} (\dot{n}_5 - \dot{n}_6)$$

$$P: \dot{n}_{P5} + \dot{n}_{P6} = \dot{n}_{P4}$$

$$x_{P5} \dot{n}_5 + x_{P6} \dot{n}_6 = x_{P4} \dot{n}_4$$

$$Q: \dot{n}_{Q5} + \dot{n}_{Q6} = \dot{n}_{Q4}$$

$$x_{Q5} \dot{n}_5 + x_{Q6} \dot{n}_6 = x_{Q4} \dot{n}_4$$

reactor balance:

$$A: n_{3A} = n_{1A} - 2\dot{f}_1$$

$$B: n_{3B} = n_{2B} - \dot{f}_1 - \dot{f}_2$$

$$Q: n_{3Q} = n_{2Q} + \dot{f}_2$$

$$P: n_{3P} = n_{1P} + \dot{f}_1$$

~~2~~ 2

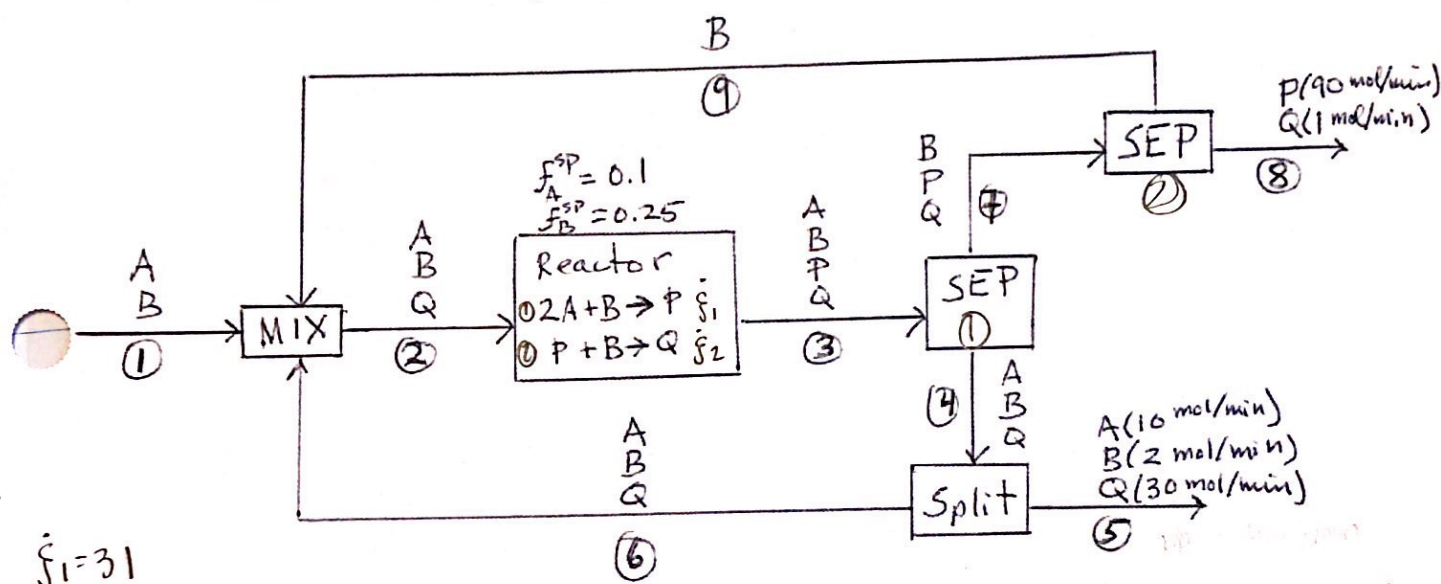
$$A: 0.1 =$$

26

(b)

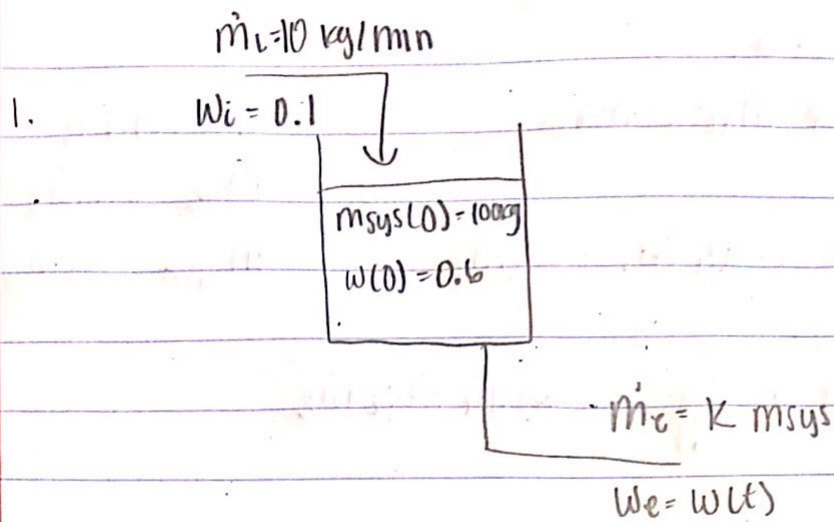
(2) Consider the block flow diagram below. Notice that the streams are labeled 1 through 9. Species molar flow rates are given by the species symbol with the stream number as a subscript. The fresh feed contains species A and B. This stream is mixed with two recycle streams, one containing A, B and Q and the other only B. The mixture enters a reactor where two reactions take place: $2A + B \rightarrow P$ and $P + B \rightarrow Q$. The reactor converts 10% of A and 25% of B. Species P is the desired product and species Q is a byproduct. The outlet stream from the reactor goes to the first separator. The top stream from the first separator goes to a second separator which has a bottom stream, the product stream, containing P and Q with the species molar flow rate of $P_8 = 90 \text{ molP/min}$ and $Q_8 = 1 \text{ molQ/min}$. The second separator's top stream is pure B and is recycled to the reactor. The bottom stream from the first separator contains A, B, and Q which is sent to a splitter which purges some of the stream and recycles the rest. The purge stream has species flow rates of $A_5 = 10 \text{ molA/min}$, $B_5 = 2 \text{ molB/min}$, and $Q_5 = 30 \text{ molQ/min}$.

- (a) Find all of the species molar flow rates not given in the problem statement. Fill in the table below. (40 pt.)
- (b) Calculate the fractional selectivity and fractional yield based on species B for both a single pass through the reactor and for the overall process using the definitions on page 299 of Murphy (10 pts.)



In addition to your calculations, fill in this table with the appropriate species flow rates in mole/min to receive full credit. Either submit an image of this table completed, or reproduce the table on your own paper and fill it in. Be sure to include the completed table with your PDF submission of your calculations.

↓/→	1	2	3	4	5	6	7	8	9
A					10		---	---	---
B					2			---	
P	---	---	90 $\frac{\text{mol}}{\text{min}}$	---	---	---	90 $\frac{\text{mol}}{\text{min}}$	90	---
Q	---				30		1 $\frac{\text{mol}}{\text{min}}$	1	---



$$\frac{dm_{\text{sys}}}{dt} = \dot{m}_i - \dot{m}_e = 10 - K m_{\text{sys}}$$

$$10 - 0.15 m_{\text{sys}}$$

$$\frac{dm_{\text{sys}}}{dt} = 10 - K m_{\text{sys}}$$

$$\int_{100}^m \frac{1}{10 - K m_{\text{sys}}} dm = \int_0^t 1 dt$$

$$\int_{100}^m \frac{1}{10 - .15 m_{\text{sys}}} dm_{\text{sys}} \quad u = 10 - .15 m \quad \frac{-1}{.15} \ln |u|$$

$$\left. -\frac{1}{.15} \ln |10 - .15 m_{\text{sys}}| \right|_{100}^m = t$$

evaluate at lower limit

$$-\frac{1}{.15} \ln |10 - .15 m_{\text{sys}}| = t$$

$$b) \frac{10 - e^{-.15(10)}}{.15} = m_{\text{sys}}$$

$$e^{-.15t} = 10 - .15 m_{\text{sys}}$$

$$10 - e^{-.15t} = .15 m_{\text{sys}}$$

$$\frac{10 - e^{-.15t}}{.15} = m_{\text{sys}}$$

$$m_{\text{sys}} = 65.17 \text{ kg}$$

balance on A:

$$c) \frac{dm_{sys}}{dt} = \dot{m}_{Ai} - \dot{m}_{Ae}$$

$$\dot{m}_{Ai} = w_i \dot{m}_i$$

$$\dot{m}_{Ae} = w_e \dot{m}_e = w(0.15 \dot{m})$$

$$m_{sysA} = w_A m_{tot}$$

$$\frac{d(wm)}{dt} = w_i \dot{m}_i - w_e \dot{m}_e$$

$$w \frac{dm}{dt} + m \frac{dw}{dt} = w_i \dot{m}_i + w_e \dot{m}_e$$

$$10 \cdot 0.15 (65.17)$$

$$= 22$$

$$w(12) + m \frac{dw}{dt} = 0.6(10) -$$

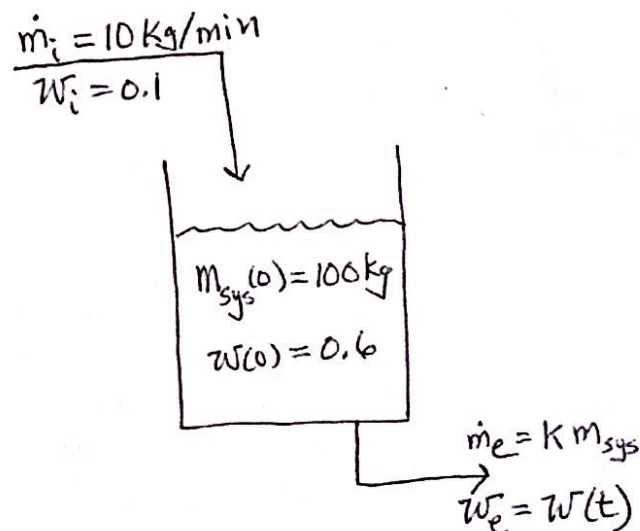
$$t=0: \frac{10 - e^0}{.15} \cdot m_{sys}$$

29

To receive full credit, you must show all of your work.

- (1) Consider a binary liquid mixture of species A and B. A storage tank initially contains 100 kg of a 60 wt% of A in the mixture. That is, the weight fraction of the A in the tank is initially $w = 0.6$. Starting at time $t = 0$, a 10 wt% of A, $w_i = 0.1$, is added at a constant rate of $\dot{m}_i = 10 \text{ kg/min}$ while at the same time the exit flow rate is proportional to the current mass in the tank such that $\dot{m}_e = k m_{\text{sys}}$. The constant $k = 0.15 \text{ min}^{-1}$ so that the initial exit flow rate is $\dot{m}_e(0) = 15 \text{ kg/min}$. Assume that the tank is well mixed so that the A weight fraction at the exit is the same as the weight fraction in the tank, $w_e = w(t)$. (50 pts.)
- (a) Derive an expression for the total mass of the mixture $m_{\text{sys}}(t)$ in the tank as a function of time. (20 pts.)
- (b) What is the total mass of the mixture in the tank in kg after 10 minutes? (5 pt.)
- (c) Write and solve a species mass balance on A. Use the result to find an expression for the weight fraction in the tank, $w(t)$. Calculate the weight fraction of A in the tank after the same 10 minutes of operation $w(t = 10 \text{ min})$ as in part (b). (25 pts.)

HINT: This problem differs from Quiz #3 by $\dot{m}_e$ the exit flow rate. In Quiz #3 the exit flow rate was constant while for this problem it is proportional to the mass in the tank. The unsteady balances can be written as separable first-order differential equations.



$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{1}{x} dx = \ln |x|$$

$$\int (a + bx) dx = \frac{1}{2b} (a + bx)^2$$

$$\int \frac{dx}{a + bx} = \frac{1}{b} \ln |a + bx|$$

$$\int \frac{dx}{x(a + bx)} = -\frac{1}{a} \ln \left| \frac{a + bx}{x} \right|$$

$$\int \frac{dx}{(a + bx)^2} = \frac{-1}{b(a + bx)}$$

$$\int \frac{dx}{(a + fx)(c + gx)} = -\frac{1}{ag - cf} \ln \left| \frac{c + gx}{a + fx} \right|$$